

GraphReAct: Adaptive Graph-of-Thought Reasoning for Tool-Using Agents on Complex Problem Solving

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Abstract—Large language models (LLMs) have demonstrated remarkable progress in complex reasoning tasks; however, solving problems that require deep, multi-step mathematical and scientific reasoning—such as PhD-level question answering or advanced symbolic computation—remains challenging. These difficulties become particularly pronounced when effective problem solving requires both structured reasoning and interaction with external tools, such as calculators, symbolic solvers, or retrieval systems.

Recent advances in agentic frameworks, most notably ReAct, have shown that interleaving reasoning traces with task-specific actions can significantly improve performance by enabling LLMs to iteratively reason, act, and observe. In ReAct, the reasoning component is typically implemented using linear Chain-of-Thought (CoT) prompting, where intermediate thoughts follow a single sequential path. While effective for many tasks, such linear reasoning can be restrictive for complex problems that naturally branch into multiple competing hypotheses, intermediate solution paths, or tool-dependent subproblems.

To address this limitation, we propose GraphReAct, a hybrid agentic framework that replaces the linear Chain-of-Thought reasoning in ReAct with an *Adaptive Graph-of-Thought* (AGoT) mechanism. AGoT dynamically decomposes a problem into interdependent subproblems organized as a directed acyclic graph, allowing the model to explore, evaluate, and prune multiple reasoning branches in parallel. Only subproblems that require further analysis are expanded, enabling more flexible and efficient exploration of complex solution spaces.

Within GraphReAct, this graph-structured reasoning is seamlessly integrated into the standard ReAct loop, preserving its action and observation stages for tool use. After identifying the most promising reasoning nodes, the agent selectively invokes external tools to validate intermediate results, perform precise computations, or retrieve supporting information, and incorporates the observations back into the reasoning graph.

We evaluate GraphReAct on challenging benchmarks, including GPQA Diamond and HARDMath Level 5, using multiple open-weight LLMs. We further compare GraphReAct against standalone prompting strategies such as Chain-of-Thought, Graph-of-Thought, Adaptive Graph-of-Thought, AIoT, and the original ReAct framework. Our analysis suggests that combining adaptive graph-structured reasoning with tool-augmented agents provides a more effective paradigm for complex problem solving than linear reasoning alone, highlighting the importance of reasoning topology in agentic LLM systems.

Index Terms—Large Language Models, Agentic AI, ReAct, Adaptive Graph-of-Thought, Graph-Based Reasoning, Tool-Augmented Reasoning

I. INTRODUCTION

Large Language Models (LLMs) have achieved notable success across a wide range of natural language and reasoning tasks. However, benchmarks that require deep scientific understanding or advanced mathematical reasoning continue to expose fundamental limitations in current models. Tasks such as PhD-level question answering and high-difficulty mathematical problem solving often demand long-horizon reasoning, hypothesis tracking, and precise verification—capabilities that remain fragile in many LLM-based systems.

A key contributor to these failures is the reliance on *linear* reasoning structures. Chain-of-Thought (CoT) [1] prompting improves performance by encouraging intermediate reasoning steps, but enforces a single sequential trajectory. For complex problems with branching solution paths, early incorrect assumptions can propagate through the reasoning chain, leading to irrecoverable errors.

Agentic frameworks have emerged as a promising direction to mitigate these issues. ReAct interleaves reasoning with actions and observations, enabling LLMs to use external tools such as calculators or symbolic solvers during problem solving. [2] Despite its success, ReAct continues to rely on linear Chain-of-Thought reasoning internally, limiting its ability to explore alternative hypotheses or recover from early mistakes.

In this work, we propose **GraphReAct**, a hybrid agentic framework that replaces linear Chain-of-Thought reasoning in ReAct with *Adaptive Graph-of-Thought* (AGoT) [3] reasoning. By restructuring the reasoning process as an adaptive graph rather than a chain, GraphReAct enables parallel exploration and selective pruning of reasoning paths while preserving ReAct’s tool-using capabilities.

Contributions:

- We introduce GraphReAct, a hybrid framework integrating AGoT reasoning into the ReAct architecture.
- We provide a controlled comparison across CoT, GoT [4], AGoT, AIoT, ReAct, and GraphReAct.
- We evaluate the impact of reasoning topology on complex problem solving using GPQA Diamond and HARDMath.

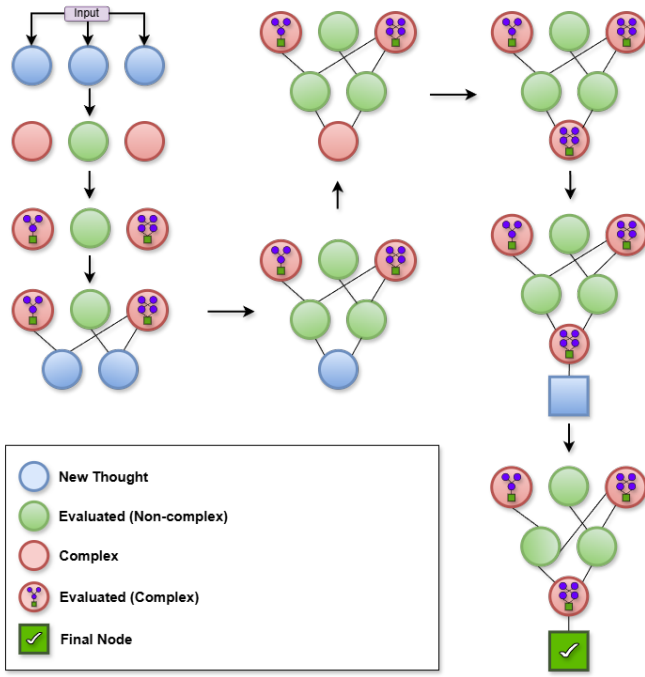


Fig. 1. Adaptive Graph-of-Thought (AGoT) reasoning. Multiple hypotheses are explored in parallel using a dynamically expanding and pruned reasoning graph.

II. RELATED WORK

Chain-of-Thought prompting demonstrated that explicitly modeling intermediate reasoning steps improves LLM performance on arithmetic and logical tasks. Subsequent work proposed non-linear extensions such as Graph-of-Thought and Adaptive Graph-of-Thought, enabling parallel reasoning and dynamic expansion.

ReAct introduced a framework for tool-augmented reasoning by interleaving thought and action steps. While effective, ReAct does not modify the internal structure of reasoning, retaining linear CoT. Other agentic approaches such as multi-agent debate focus on interaction rather than reasoning topology. [5], [6]

GraphReAct addresses this gap by unifying adaptive graph-based reasoning with tool-using agents.

III. METHODOLOGY

A. Overview

GraphReAct is a hybrid agentic framework that integrates *Adaptive Graph-of-Thought* (AGoT) reasoning into the ReAct paradigm. Unlike standard ReAct, which relies on linear Chain-of-Thought reasoning, GraphReAct structures the reasoning process as an adaptive directed acyclic graph (DAG), enabling parallel exploration of multiple hypotheses while preserving the action–observation loop for tool usage.

The framework operates in three stages: (1) adaptive graph-based reasoning, (2) selective tool interaction via ReAct, and (3) iterative refinement until convergence.



Fig. 2. Linear Chain-of-Thought (CoT) reasoning. The model follows a single sequential reasoning path, making it vulnerable to early errors that propagate through the entire solution.

B. Adaptive Graph-of-Thought Reasoning

Given an input problem, GraphReAct initializes a reasoning graph where each node represents a partial hypothesis, intermediate derivation, or subproblem. Edges encode logical or computational dependencies between nodes.

At each iteration, the agent evaluates existing nodes and selectively expands those with high uncertainty or unresolved dependencies. This adaptive expansion avoids exhaustive branching and limits graph growth to reasoning paths that are likely to contribute to the final solution. Nodes that are inconsistent, redundant, or contradicted by evidence are pruned from the graph.

By maintaining multiple active reasoning paths, AGoT reduces sensitivity to early errors that commonly affect linear Chain-of-Thought reasoning.

C. Integration with ReAct

Once a subset of promising reasoning nodes is identified, GraphReAct integrates the ReAct action–observation loop. For each selected node, the agent decides whether external tool usage is required, such as invoking a calculator, symbolic solver, or retrieval function.

Tool invocations are executed as ReAct actions, and their outputs are treated as observations. These observations are then injected back into the reasoning graph, where they are used to validate hypotheses, resolve dependencies, or prune incorrect branches.

This selective tool usage ensures that external resources are applied only when necessary, improving both accuracy and efficiency.

D. Iterative Refinement and Convergence

The reasoning graph is iteratively updated through alternating reasoning and action phases until a convergence criterion is met. Convergence is defined as the emergence of a stable solution node that is internally consistent and supported by tool-based observations.

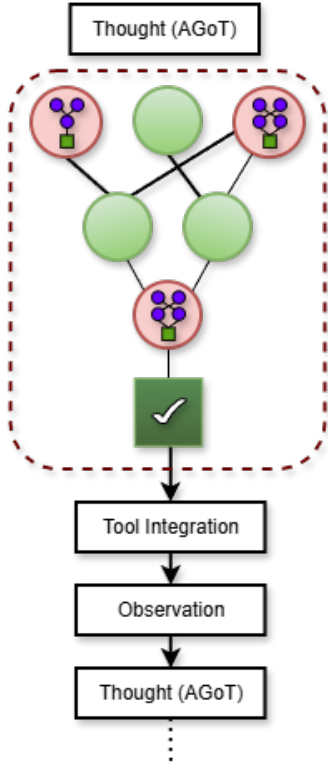


Fig. 3. GraphReAct framework integrating Adaptive Graph-of-Thought reasoning with the ReAct action–observation loop.

The final answer is extracted from the highest-confidence leaf node in the graph, representing the most coherent and validated reasoning path.

IV. EXPERIMENTAL SETUP

A. Models

We conduct experiments using two open-weight large language models:

- **DeepSeek-Math-7B-Instruct**, a math-focused instruction-tuned model.
- **Qwen2-7B**, a general-purpose instruction-tuned model.

Both models are evaluated under identical decoding parameters and tool configurations to ensure fair comparison across reasoning methods.

B. Dataset

Evaluation is performed on the **GPQA Diamond** benchmark, [7] which consists of 198 PhD-level multiple-choice questions spanning physics, chemistry, biology, astronomy, and related scientific domains. GPQA Diamond is designed to test deep reasoning rather than surface-level pattern matching.

C. Baselines

We compare GraphReAct against the following reasoning paradigms:

- **Chain-of-Thought (CoT)**: Linear reasoning with explicit intermediate steps.

- **Graph-of-Thought (GoT)**: Non-adaptive graph-based reasoning without pruning.
- **Adaptive Graph-of-Thought (AGoT)**: Graph-based reasoning with dynamic expansion and pruning.
- **ReAct**: Linear Chain-of-Thought interleaved with actions and observations.

All baselines use the same prompt budget and access to identical tools where applicable.

D. Evaluation Metrics

Performance is measured using *exact-match accuracy*, computed as the percentage of correctly answered questions. We additionally report reasoning efficiency in terms of the average number of reasoning steps per problem for agentic methods.

E. Implementation Details

All experiments are conducted under controlled conditions with fixed random seeds. Each question is evaluated independently, and no cross-question memory is used. For agentic methods, the maximum number of reasoning steps per question is capped to prevent runaway computation.

Tool usage is restricted to deterministic functions to ensure reproducibility. No external fine-tuning or task-specific training is performed.

V. RESULTS

We evaluate GraphReAct (AGoT-ReAct) against five reasoning paradigms on the **GPQA Diamond** benchmark using DeepSeek-Math-7B-Instruct and Qwen2-7B. Performance is measured using exact-match accuracy over 198 PhD-level scientific questions.

A. Overall Accuracy

Table I presents a comprehensive comparison of all reasoning paradigms on the GPQA Diamond benchmark. GraphReAct achieves the **highest average accuracy** across both models (31.91%), outperforming linear Chain-of-Thought reasoning, standalone graph-based methods, and the original ReAct framework.

On **Qwen2-7B**, GraphReAct attains **35.03% accuracy**, exceeding ReAct by 1.70 percentage points and improving substantially over pure Adaptive Graph-of-Thought (+8.08%). This indicates a strong synergy between adaptive graph reasoning and tool-augmented action for this model. In contrast, on **DeepSeek**, pure AGoT yields the highest accuracy (30.33%), while GraphReAct remains competitive and improves over ReAct by 3.03%. These results suggest that the effectiveness of hybrid reasoning–tool integration is model-dependent.

Overall, the results in Table I demonstrate that while graph-based reasoning generally outperforms linear reasoning, combining adaptive graph reasoning with selective tool usage yields the most robust performance across heterogeneous models.

TABLE I
COMPREHENSIVE ACCURACY COMPARISON ON GPQA DIAMOND ACROSS REASONING METHODS AND MODELS

Method	DeepSeek-Math-7B	Qwen2-7B	Average
Chain-of-Thought (CoT)	29.29%	28.78%	29.04%
Graph-of-Thought (GoT)	28.79%	32.32%	30.56%
Adaptive Graph-of-Thought (AGoT)	30.33%	26.95%	28.64%
ReAct	25.76%	33.33%	29.55%
GraphReAct (AGoT-ReAct)	28.79%	35.03%	31.91%

B. Efficiency Analysis

Accuracy alone does not fully characterize the effectiveness of agentic reasoning systems. As shown in Table II, GraphReAct substantially reduces the computational effort required to reach a solution. Compared to ReAct, GraphReAct reduces the average number of reasoning steps per problem by 24.0% on DeepSeek (5.0 $\rightarrow$ 3.8) and by 56.8% on Qwen2-7B (4.4 $\rightarrow$ 1.9).

These reductions indicate that adaptive graph pruning enables the agent to avoid redundant or unproductive reasoning paths and invoke external tools only when they are most informative. Importantly, these efficiency gains are achieved while maintaining or improving accuracy.

C. ReAct vs. GraphReAct

To isolate the impact of replacing linear Chain-of-Thought reasoning within the ReAct framework, we directly compare ReAct and GraphReAct in Table II.

TABLE II
REACT VS. GRAPHREACT COMPARISON ON GPQA DIAMOND

Model	Method	Accuracy	Avg. Steps	Total Steps
DeepSeek	ReAct	25.76%	5.0	983
	GraphReAct	28.79%	3.8	754
Qwen2	ReAct	33.33%	4.4	869
	GraphReAct	35.03%	1.9	381

Table II shows that GraphReAct consistently improves both accuracy and efficiency over ReAct across models. On DeepSeek, GraphReAct answers six additional questions correctly while reducing the total number of reasoning steps by 23.3%. On Qwen2-7B, GraphReAct achieves higher accuracy using less than half the total reasoning steps, highlighting a strong accuracy–efficiency trade-off.

Taken together, the results in Tables I and II confirm that the performance gains of GraphReAct arise from integrating adaptive graph-structured reasoning with selective tool usage, rather than from increased computation alone. This underscores the importance of reasoning topology in the design of effective tool-using language agents.

VI. DISCUSSION

The experimental results emphasize the importance of reasoning topology in tool-augmented language agents. Linear Chain-of-Thought reasoning constrains the agent to a single hypothesis trajectory, making it vulnerable to early errors. In

contrast, graph-based reasoning enables parallel exploration and selective pruning of competing hypotheses.

GraphReAct further extends this advantage by integrating adaptive graph reasoning with the ReAct action–observation loop. This integration allows the agent to invoke external tools only for promising reasoning nodes, leading to both improved accuracy and significant reductions in reasoning steps.

The observed model-dependent behavior suggests that architectural characteristics of the underlying LLM influence how effectively tool usage interacts with structured reasoning. While DeepSeek benefits most from pure adaptive graph reasoning, Qwen2 shows strong synergy between graph reasoning and tool interaction, achieving the highest overall performance.

VII. LIMITATIONS

GraphReAct introduces additional design complexity compared to linear reasoning frameworks, including the need for node expansion, scoring, and pruning strategies. Although the framework improves efficiency in practice, maintaining and updating a reasoning graph incurs overhead that may affect scalability for very large graphs.

Furthermore, the effectiveness of GraphReAct is model-dependent. While substantial gains are observed on Qwen2-7B, improvements on DeepSeek are more modest, indicating that hybrid reasoning–tool frameworks may require model-specific tuning.

Finally, our evaluation is limited to the GPQA Diamond benchmark. While GPQA provides a rigorous test of scientific reasoning, future work should assess generalization to other domains and task formats.

VIII. CONCLUSION

We introduced **GraphReAct**, a hybrid agentic framework that integrates Adaptive Graph-of-Thought reasoning into the ReAct architecture. By replacing linear Chain-of-Thought reasoning with an adaptive graph structure, GraphReAct enables parallel hypothesis exploration, selective tool usage, and efficient pruning of unpromising reasoning paths.

Empirical results on the GPQA Diamond benchmark demonstrate that GraphReAct achieves the best overall accuracy across models while significantly reducing reasoning steps, particularly on Qwen2-7B. These findings highlight that reasoning topology is a critical factor in the design of tool-using LLM agents and that adaptive graph-based inference provides a robust foundation for complex problem solving.

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